

Worksheet — 1.3 Exchanging Data

Name: _____ Date: _____

OCR A Level Computer Science (H446) — Component 01, Section 1.3 (Compression, encryption & hashing; Databases; Networks; Web technologies)

Time: ~35 min.

1. Key-term match

Match each term (A–H) to its definition (1–8). Write the letter next to the number.

	Term		Definition
A	Lossy compression	1	Translates domain names into IP addresses
B	Hashing	2	A foreign key must reference a valid existing primary key
C	RLE	3	One-way function producing a fixed-length digest
D	DNS	4	Permanently removes data; original cannot be recovered
E	Referential integrity	5	Public + private key pair used for encryption
F	Asymmetric encryption	6	Replaces runs of repeated values with value + count
G	Primary key	7	Organising data to reduce redundancy & anomalies
H	Normalisation	8	Attribute(s) that uniquely identify each record

Answers: A- B- C- D- E- F- G- H-

2. Run Length Encoding (RLE)

(a) Encode the following pixel run as RLE pairs of (value, count):

R R R R G G B B B W W W W

Working space:

(b) Decode this RLE data back into the original sequence. Each pair is (value, count):

(3,A) (1,B) (4,A) (2,C)

Working space:

(c) Give **one** type of data for which RLE would *increase* the file size, and say why.

3. Symmetric vs asymmetric encryption — fill in

Complete the blanks:

a) **Symmetric** encryption uses the ___ key to encrypt and decrypt. Its main weakness is the ___ problem.

b) **Asymmetric** encryption uses a ___ key (shared openly) and a ___ key (kept secret).

c) Data encrypted with someone's **public** key can only be decrypted with their ___ key.

d) Asymmetric is ___ (faster / slower) than symmetric.

e) HTTPS uses asymmetric encryption to securely exchange a ___ session key, then uses ___ encryption for the bulk data because it is faster.

4. Normalisation

An unnormalised `STUDENT_COURSES` table is shown below.

StudentID	StudentName	Courses	TutorID	TutorName
S01	Amy Khan	Maths, Physics	T1	Mr Bell
S02	Ben Cole	Maths	T1	Mr Bell
S03	Cara Lee	Physics, Chemistry	T2	Ms Frost

(a) State **one** repeating group that breaks **1NF**, and explain how to fix it.

(b) The `TutorName` depends on `TutorID`, not on the student. What type of dependency is this, and which normal form does it break?

(c) Redesign the data into **3NF**. Sketch the tables (table name + columns). Underline the **primary key** in each and mark any **foreign key** with (FK).

Table 1: _____ (_____)

Table 2: _____ (_____)

Table 3: _____ (_____)

Table 4 (if needed): _____ (_____)

5. Write the SQL

Use these tables: `Student(StudentID, Name, TutorID, YearGroup)` and `Tutor(TutorID, TutorName)`.

(a) Return the **Name** and **YearGroup** of every student in Year 12, sorted by Name (A–Z).

(b) Return each student's **Name** alongside their **TutorName** (join the two tables on TutorID).

(c) Change the **YearGroup** to 13 for the student whose StudentID is `S01`.

6. TCP/IP layer matching

Write each protocol in the row of the layer it belongs to.

Protocols: `HTTP`, `TCP`, `IP`, `Ethernet`, `SMTP`, `UDP`, `Wi-Fi`, `FTP`

TCP/IP layer	Protocol(s)
Application	_____
Transport	_____
Network / Internet	_____
Link / Data link	_____

7. Short answer

(a) In **packet switching**, give two things that are added to each packet so it can travel the network and be reassembled.

(b) What does **DNS** do, and why is caching used?

(c) In **PageRank**, why is a link from a high-ranked page worth more than a link from a low-ranked page?

(d) Give one reason why login/payment checks must run **server-side** rather than only client-side.

Challenge box

A website stores user passwords. Explain why the passwords should be **hashed** rather than **encrypted**, and describe what happens when a user later logs in (refer to *one-way*, *deterministic*, and *collision*).

Answer key

1. **Key-term match** A-4, B-3, C-6, D-1, E-2, F-5, G-8, H-7

2. **RLE** - (a) (R,4) (G,2) (B,3) (W,5) — i.e. 4 R, 2 G, 3 B, 5 W. - (b) A A A B A A A A C C (3 A, 1 B, 4 A, 2 C). - (c) Data with **few/no runs** of repeats — e.g. a detailed photograph or already-compressed text. Each unique value still needs a count, so the file *grows* (no repetition to exploit). RLE is lossless and only shrinks files with long runs.

3. **Symmetric vs asymmetric** - (a) **same** key; **key distribution / key exchange** problem. - (b) **public** key; **private** key. - (c) **private** key. - (d) **slower**. - (e) **symmetric** session key; then **symmetric** encryption for bulk data.

4. **Normalisation** - (a) The **Courses** column holds multiple values ("Maths, Physics") — a **repeating group** / non-atomic value. To reach 1NF, make values atomic: one course per row (or move courses to a separate table) and ensure a suitable primary key. - (b) **Transitive dependency** (TutorName depends on TutorID, a non-key attribute, not directly on the key). This breaks **3NF**. (Partial dependencies break 2NF; transitive break 3NF.) - (c) Sample 3NF design (primary keys in **bold**, foreign keys marked FK): -
Student(**StudentID** , Name, TutorID (FK)) - Tutor(**TutorID** , TutorName) - Course(**CourseID** , CourseName) - StudentCourse(**StudentID (FK), CourseID (FK)**) — linking/junction table with a **composite** primary key to resolve the many-to-many between students and courses.

5. **SQL** (accept reasonable variations / casing)

```
-- (a)
SELECT Name, YearGroup
FROM Student
WHERE YearGroup = 12
ORDER BY Name ASC;

-- (b)
SELECT Student.Name, Tutor.TutorName
FROM Student
JOIN Tutor ON Student.TutorID = Tutor.TutorID;

-- (c)
UPDATE Student
SET YearGroup = 13
WHERE StudentID = 'S01';
```

6. TCP/IP layer matching - Application: **HTTP, SMTP, FTP** - Transport: **TCP, UDP** - Network / Internet: **IP**
- Link / Data link: **Ethernet, Wi-Fi**

7. Short answer - (a) Any two of: **source IP address, destination IP address, a sequence/packet number** (for reassembly in the right order). (Also acceptable: error-check/checksum, TTL.) - (b) **DNS translates a domain name (e.g. example.com) into an IP address** so the device can locate the server. Caching stores recently looked-up results, so repeat lookups are **faster** and reduce traffic/load on DNS servers. - (c) PageRank treats each incoming link as a "vote" of importance; a high-ranked page passes on **more of its rank** through its links, so its vote counts for more. (Damping factor ~0.85; iterates until values converge.) - (d) Client-side checks can be **viewed, bypassed or disabled** by the user (e.g. editing the page / disabling JavaScript), so they are not secure. Server-side checks are hidden and authoritative, so security-critical checks must run there.

Challenge box (model points) - Hashing is **one-way**: even if the database is stolen, attackers cannot reverse the hash back to the password (unlike encryption, which is reversible with a key). So the plaintext password is never stored. - It is **deterministic**: the same password always produces the same hash. On login, the entered password is hashed and **compared to the stored hash** — if they match, access is granted; the password itself is never compared. - A good hash function has **few collisions** (different inputs rarely produce the same hash), so a wrong password is very unlikely to produce a matching hash. (Real systems also add a *salt*.)